Homework2

PS 2010

Due: Sep 26th, 9:29:59am

Notice: The grading will mainly focus on the steps not the final results. This means even your final results are wrong but you follow the correct steps, you will be given the full score.

1 Computing Indefinite Integral (10 Points)

- Determine $f(x)$ given $f^{'}(x) = 6x^8 - 20x^4 + x^2 + 9$
- $\int 12dx$
- $\int x^3 - \frac{\mathbf{e}^{-x}-4}{\mathbf{e}^{-x}}dx$
- $\int 4\mathbf{e}^z + 15 - \frac{1}{6z}dz$
- $\int (8x-12)\left(4x^2-12x\right)^4 dx$
  Hint: Integration by substitution
- $\int \frac{3x}{1+9x^2}dx$
- $\int \left(7x - 2x^3\right)\mathbf{e}^{x^4-7x^2}dx$
- $\int x^2 \ln|x|dx$

2 Computing the definite Integral

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- $\int_6^0 (2+5x)\mathbf{e}^{\frac{1}{3}x}dx$
  Hint: Integral by parts
- $\int_1^3 3\sqrt{x}dx$

3 R Question

Review the method we learned in the class on how to approximate area using Monte Carlo Integration. Now, let's change our function to a normal distribution function and calculate the area within a interval. You need to do:

- Random draw some data from normal distribution using code:
- $x_value < -rnorm(10000)$
- $y_value < -dnorm(x_value)$

- In the above code function $dnorm()$ will directly gives you the output values for every value in x_value. Remember this point, you need to use it later.
- Q1: Draw a line plot where x-axis is x_vlaue and y-axis is y_value
- Q2: Then, try to plot all data to the above line plot. Your plot should looks similar to the plot in Figure 1 below. You could borrow code from the R lecture.
- Q2: Now, using the knowledge we learned from class, can you approximate the area between interval [-2, 2] using Monte Carlo Integration. What is area?
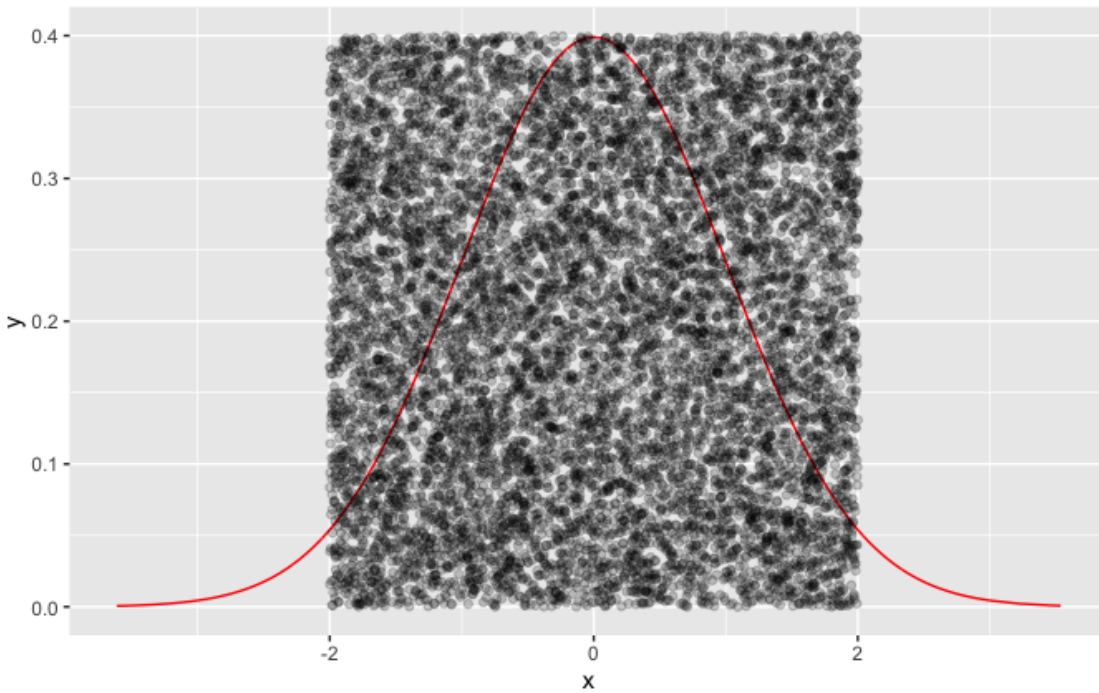
Figure 1: